

Day 09—Linux-Basics

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THEORY

1. What is Linux?

It is an open source operating system(open and free to use and distribute) family widely used in servers ,networking , and cybersecurity, available in distributions like parrot os and kali linux, its traditionally faster and more secure.

2. What is a Linux distribution?

It is a version of linux that combines linux kernel with software, tools and a package manager to create an operating system.

3. Name four Linux distributions.

Ubuntu , kali linux, parrot os and fedora

4. What is Kali Linux?

It is a free linux distribution designed for security testing and cybersecurity work.

5. What is a terminal?

It is where we communicate with linux by commands, it is faster than using gui.

6. What does the whoami command do?

It shows the username of the logged in user

7. What does hostname display?

Computer's host name

8. What does pwd mean?

Print working directory , it shows where we are, giving a full path.

9. What is the purpose of ls?

It lists the content in the directory.

10. Explain the difference between ls, ls -l and ls -a.

Ls: lists files and directories

ls -l: lists files with permissions ,owner,size and date.

Ls -a: used to list all files even hidden ones

11. What does cd do?

Change directory:it let us move to the directory we want

12. What does cd .. do?

We go back to the directory we were in.

13. What does cd ~ do?

Moves to the current user home directory

14. What does cd / do?

We go to the root directory.

15. What is the Linux root directory?

It is (/)

16. What is the difference between the root directory and the root user?

The root directory is the file system root directory (/) the top of the tree

While the root user is the user that has privileged administrative access

17. What is an absolute path?

It is the complete path starting from the filesystem root.

18. What is a relative path?

It depends on where you stand

19. How are hidden Linux files commonly identified?

By ls -a

20. Why are Linux skills important for cybersecurity?

Linux is essential in cybersecurity for labs, servers, monitoring, penetration testing, forensics, and DevOps. These skills support tools like Kali Linux, Nmap, Wireshark, and Burp Suite.

PRACTICAL SCREENSHOTS

whoami · hostname · pwd · ls · ls -l · ls -a · cd / · cd ~

I used kali linux for this task

```
kali@kali: /
Session Actions Edit View Help
(kali@kali)-[~]
$ whpami
Command 'whpami' not found, did you mean:
  command 'whoami' from deb coreutils
Try: sudo apt install <deb name>

(kali@kali)-[~]
$ whoami
kali

(kali@kali)-[~]
$ hostname
kali

(kali@kali)-[~]
$ pwd
/home/kali

(kali@kali)-[~]
$ ls
Desktop  Downloads  Pictures  Public  Videos
Documents Music      Projects  Templates

(kali@kali)-[~]
$ ls -l
total 36
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Desktop
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Documents
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Downloads
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Music
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Pictures
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Projects
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Public
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Templates
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Videos

(kali@kali)-[~]
$ ls -la
.          Projects
..         Public
.bash_logout .ssh
.bashrc    Templates
.bashrc.original .vboxclient-clipboard-tty7-control.pid
.cache     .vboxclient-clipboard-tty7-service.pid
.config    .vboxclient-draganddrop-tty7-control.pid
Desktop    .vboxclient-draganddrop-tty7-service.pid
.dmrc      .vboxclient-hostversion-tty7-control.pid
Documents  .vboxclient-seamless-tty7-control.pid
Downloads  .vboxclient-seamless-tty7-service.pid
.face      .vboxclient-vmvga-session-tty7-control.pid
.face.icon .vboxclient-vmvga-session-tty7-service.pid
.gnupg     Videos
.ICEauthority .Xauthority
.java      .xsession-errors
.local     .xsession-errors.old
.Music     .zprofile
.Pictures  .zsh_history
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